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Batch/Section: 28th_A(day)

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Department name: Computer Science & Engineering

Faculty Name: Muhammad Anwarul Azim

University name: University of Chittagong.

Course No.: CSE 423

Course Title: Simulation & Modeling.

Course Credits: 3.00

Assignment No.: 03

Online Google Form Submission Date: 05 May,2026

Total Count of Students in this course/class/batch: 61

Total Count of Exercises (or questions or problems): 197

Chapter No. : 07

Chapter Name: Random-Number Generation

The problem (question) serial number: 122

The problem (question) Description: Discuss the types of auto-correlation of random numbers.

Answers: (122) Autocorrelation: Autocorrelation tests are concerned with the dependence between numbers in a sequence. The test requires computation of autocorrelation between every m numbers, here m is called the lag, starting with the i th numbers.

So, the autocorrelation ρ_{im} is computed between:

$$R_i, R_{i+m}, R_{i+2m}, \dots, R_{i+(M+1)m}$$

where M is the largest integers such that:

$$i + (M+1)m \leq N$$

Types of Autocorrelation of random numbers:

① Positive Autocorrelation:

A high value is likely to be followed by another high value, and a low value by a low value. It will occur when $\rho_{im} > 0$.

Successive values at lag m have a higher prob. than expected of being close in value.

Example: 0.85, 0.79, 0.91 all high values following each other.

② Negative Autocorrelation:

A high value is likely to be followed by another high value, and a low value by a low value. Occurs when $\rho_{im} < 0$. Low random numbers tend to be followed by high ones.

High random numbers to be followed by low ones. Values alternate between high and low. Example: 0.12, 0.89, 0.08, 0.91 alternating low and high values.

③ Zero Co Autocorrelation:

There is absolutely no linear relationship between a value in a sequence and its past or future versions. It occurs when $\rho_{im} = 0$. No discernible relationship between successive random numbers at lag m . Random numbers are truly independent of each other.

Example: 0.23, 0.87, 0.45, 0.61 $\rightarrow$ no pattern, completely random.

Understanding the types of autocorrelation is vital for verifying that a sequence of random numbers is truly independent and unpredictable. By identifying positive, negative, zero autocorrelation, researchers can ensure their simulation are free from hidden patterns and statically sound.